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MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-BCA (HONOURS) REGULAR EXAMINATION March 2026 Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECBCA202 - DESIGN THINKING &INNOVATION

(2024 ADMISSION ONWARDS)

SCHEME FOR EVALUATION

Duration : 1 Hour **Max.Marks :35**

Part-A

Short Answer Questions (7 out of 10 Questions) Each questions carries 5 marks.

1. Discuss techniques to gather customer insights.

   Definition (1 mark) Customer insights are the understanding of customer needs, behaviours, and problems.

   Techniques (Any 4 points – 1 mark each)

   Interviews
   Direct interaction with customers to understand feelings and needs.
   Observation
   Watching users while using a product or service.
   Surveys and Questionnaires
   Collecting opinions from many users quickly.
   Focus Groups
   Group discussion to gather different viewpoints.
   Customer Journey Mapping
   Studying customer experience step by step.
   Feedback Analysis
   Examining reviews and complaints.

2. Explain the concept of creative thinking.

   Definition (2 marks) Creative thinking is the ability to generate new and innovative ideas to solve problems.

   Features (Any 3 points – 1 mark each)

   Originality
   Imagination
   Flexibility
   Problem-solving ability
   Innovation
   Example:(1 mark)

3. Explain the Empathize stage with a real-life example.

   Definition (2 marks) Empathize is the first stage in design thinking where designers understand users' feelings, needs, and challenges.

   Activities (2 marks)
   Observation
   Interviews
   Interaction with users
   Real-Life Example (1 mark)

4. Define the term "Wicked Problem".

   Definition (2 marks) A wicked problem is a complex social or cultural problem with no single correct solution.

   Characteristics (Any 3 points – 1 mark each)

   Difficult to define

   Multiple causes

   No permanent solution

   Different opinions from stakeholders

   Example : (1 mark)(Traffic congestion or climate change.)

5. Explain brainstorming phases.

   Definition (1 mark) Brainstorming is a method for generating many ideas creatively.

   Explanation of phases – 4 marks

   Problem identification

   Idea generation

   Evaluation

   Selection Phases

6. Differentiate between Affective and Cognitive Empathy with Suitable Examples in Product Design.

   Affective Empathy -Focuses on feelings and emotions. Helps designers emotionally relate to users.

   Cognitive Empathy- Focuses on thoughts and perspectives. Helps designers understand user behavior and needs.

Definition 2 Marks- Differences between both-2 Marks- Suitable examples in product design-1 Mark

7. Explain the factors that influence an individual's thinking. How do these factors shape problem definition?

   Factors (Any 4 points – 1 mark each)

   Experience, Culture, Education, Emotions, Environment, Knowledge
   Problem Definition& Example (1 mark) These factors influence how individuals understand and interpret problems.

8. Discuss significance of feedback loops in design thinking.

   Definition (1 mark)Feedback loops involve continuous testing and improvement based on user feedback.

   Importance (Any 3 points – 1 mark each)

   Improves product quality, Identifies user problems early, Encourages innovation, Reduces errors

   Example (1 mark) Mobile app developers update features based on customer reviews.

9. Explain the importance of final presentation in design thinking.

   Importance (Any 5 points – 1 mark each)

   Communicates ideas clearly, Shows prototype and results, Attracts investors/stakeholders, Explains problem and solution, Builds confidence, Receives final feedback.

10. Differentiate between Low-Fidelity and High-Fidelity Prototypes.

    Low-Fidelity Prototype:

    • Simple and rough • Cheap and fast • Used in early stages

    High-Fidelity Prototype:

    • Detailed and realistic • Costly and time-consuming • Used in final stages

    Examples:

    • Low-fidelity: Paper sketch

    • High-fidelity: Interactive app demo